

HIV-1 drug resistance mutations in children after failure of first-line nonnucleoside reverse transcriptase inhibitor-based antiretroviral therapy

T Puthanakit,^{1,2,3} G Jourdain,⁴ S Hongsiriwon,⁵ P Suntarattiwong,⁶ K Choekphaibulkit,⁷ V Sirisanthana,^{2,8} P Kosalaraksa,⁹ W Petdachai,¹⁰ R Hansudewechakul,¹¹ U Siangphoe,¹ T Suwanlerk¹ and J Ananworanich^{1,12,13} on behalf of the HIV-NAT 086 study team

¹The HIV Netherlands Australia Thailand Research Collaboration (HIV-NAT), Bangkok, Thailand, ²Department of Pediatrics, Faculty of Medicine, Chulalongkorn University, Bangkok, Thailand, ³Research Institute for Health Sciences, Chiang Mai University, Chiang Mai, Thailand, ⁴Program for HIV Prevention and Treatment, Institut de Recherche pour le Développement (IRD) UMI 174, Department of Medical Technology, Faculty of Associated Medical Sciences, Chiang Mai University, Chiang Mai, Thailand, ⁵Regional Hospital, Chonburi, Thailand, ⁶Queen Sirikit National Institute of Child Health, Bangkok, Thailand, ⁷Department of Pediatrics, Faculty of Medicine, Siriraj Hospital, Mahidol University, Bangkok, Thailand, ⁸Department of Pediatrics, Faculty of Medicine, Chiang Mai University, Chiang Mai, Thailand, ⁹Department of Pediatrics, Faculty of Medicine, Khon Kaen University, Khon Kaen, Thailand, ¹⁰Petchburi Hospital, Petchburi, Thailand, ¹¹Chiang Rai Regional Hospital, Chiang Rai, Thailand, ¹²South East Asia Research Collaboration with Hawaii, Bangkok, Thailand and ¹³Department of Medicine, Faculty of Medicine, Chulalongkorn University, Bangkok, Thailand

Objectives

The aim of the study was to assess the prevalence, predictors and patterns of genotypic resistance mutations in children after failure of World Health Organization-recommended initial nonnucleoside reverse transcriptase inhibitor (NNRTI)-based treatment regimens.

Methods

We carried out a multicentre retrospective study of genotyping tests performed for all HIV-infected children at eight paediatric centres in Thailand who experienced failure of NNRTI therapy at a time when virological monitoring was not routinely available.

Results

One hundred and twenty children were included in the study. Their median age (interquartile range) was 9.1 (6.8–11.0) years, the median duration of their NNRTI regimens was 23.7 (15.7–32.6) months, their median CD4 percentage was 12% (4–20%), and their median plasma HIV RNA at the time of genotype testing was 4.8 (4.3–5.2) log₁₀ HIV-1 RNA copies/mL. The nucleoside reverse transcriptase inhibitor (NRTI) resistance mutations found were as follows: 85% of the children had M184V/I, 23% had at least four thymidine analogue mutations, 12% had the Q151M complex, 5% had K65R, and 1% had the 69 insertion. Ninety-eight per cent of the children had at least one NNRTI resistance mutation, and 48% had etravirine mutation-weighted scores ≥ 4 . CD4 percentage < 15% prior to switching regimens [odds ratio (OR) 5.49; 95% confidence interval (CI) 2.02–14.93] and plasma HIV RNA > 5 log₁₀ copies/mL (OR 2.46; 95% CI 1.04–5.82) were independent predictors of at least four thymidine analogue mutations, the Q151M complex or the 69 insertion.

Conclusions

In settings without routine viral load monitoring, second-line antiretroviral therapy regimens should be designed assuming that clinical or immunological failure is associated with high rates of multi-NRTI resistance and NNRTI resistance, including resistance to etravirine.

Keywords: children, HIV, nonnucleoside reverse transcriptase inhibitor, resistance mutations, treatment failure

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Introduction

The widespread use of antiretroviral therapy (ART) for the treatment of HIV-infected children has dramatically changed the course of HIV infection, leading to reductions in morbidity and mortality [1–3]. A triple drug combination including two nucleoside reverse transcriptase inhibitors (NRTIs) plus one nonnucleoside reverse transcriptase inhibitor (NNRTI) or one protease inhibitor (PI) [4] is widely recommended as first-line therapy. For resource-limited settings, the World Health Organization (WHO) recommends an NNRTI-based regimen, which is preferred because it is effective, well tolerated and inexpensive.

Plasma HIV RNA monitoring after initiation of ART is usually not available through treatment programmes in resource-limited settings [5]. For example, the Thai national AIDS programme provides antiretroviral drugs for HIV-infected patients and CD4 monitoring every 6 months. Annual plasma HIV viral load monitoring was only recently incorporated into the national programme in 2008. Thus, in the past, the majority of Thai children were diagnosed with treatment failure when they had clinical or immunological failure, that is a long time after virological replication had resumed while they were still on treatment. Consequently, resistance-associated mutations may have occurred in these children as a result of persistent viral replication under drug pressure.

The goal of second-line treatment is to fully suppress HIV replication; therefore, the new regimen should comprise at least two, but preferably three, fully active drugs. According to US guidelines, the recommended second-line regimen for patients who experience failure of NNRTI-based regimens is a boosted PI plus two NRTIs based on resistance testing [4]. However, in real life in resource-limited settings, it may not be feasible to perform individual resistance testing. There have been a few reports on the pattern of HIV-1 drug resistance mutations in children who experienced failure of first-line NNRTI regimens from South Africa [6], Uganda [7] and Thailand [8]. Data from an HIV-infected adult Thai cohort showed that the majority of patients who experienced failure of NNRTI regimens had M184V (89%), NNRTI resistance mutations (92%), thymidine analogue mutations (37%), Q151M (8%) and K65R (6%). High plasma HIV RNA at the time of treatment failure was associated with a higher risk of multi-NRTI resistance [9].

There is a new NNRTI, etravirine, which, in contrast to nevirapine and efavirenz, requires multiple mutations to

reduce drug susceptibility [10,11]. Therefore, it is important to assess the prevalence of etravirine-associated mutations in children who have experienced failure of first-line NNRTI treatment in order to predict the potential role of etravirine as a component of second-line regimens. The impact of mutations associated with etravirine has mainly been studied in the context of PI-based salvage regimens in adults [12].

In the present study, we aimed to describe the patterns of genotypic resistance mutations in children after failure of WHO-recommended initial NNRTI-based treatment regimens. The secondary objectives were to determine the prevalence and predictors of multidrug NRTI resistance and high-grade resistance to etravirine. The results of this study may be useful in making decisions regarding second-line antiretroviral drug regimens in children, especially in settings lacking access to individual genotypic resistance testing prior to switching to a second-line regimen, and also in planning by policy makers of the provision of second-line regimens in their national programmes.

Methods

We collected treatment outcome data from eight large paediatric HIV centres in Thailand for all children who experienced failure of NNRTI-based therapy and received a ritonavir-boosted PI regimen as second-line treatment. Following Thai national AIDS programme, monitoring after initiation of ART included clinical response and CD4 monitoring every 6 months. Plasma HIV RNA measurement was performed only when treatment failure was suspected. Treatment failure was considered to have occurred when a child showed clinical disease progression, a suboptimal immunological response, defined as an increase in the CD4 percentage of <5% or an increase in the CD4 count of <50 cells/ μ L (age >5 years) over the first year of treatment, or immunological decline, defined as a decline in the CD4 percentage of >5% or a CD4 cell count drop of >30% from peak within 6 months. Genotypic resistance testing was recommended for plasma HIV RNA >1000 copies/mL, which has been provided free of charge under the national programme since 2005.

For this HIV drug resistance pattern study, the inclusion criteria were HIV infection, age <18 years, receipt of an NNRTI-based ART regimen containing either nevirapine or efavirenz, treatment failure on first-line therapy, and had genotypic resistance testing within 12 months prior to switching to a second-line regimen. Children who had a

history of mono or dual NRTI therapy before starting NNRTI-based ART, or who received an NRTI backbone other than zidovudine plus lamivudine or stavudine plus lamivudine, were excluded from the study.

Retrospective data collection was performed using a standardized data collection form. Information obtained from medical records included patient demographics, HIV Centers for Disease Control and Prevention (CDC) clinical classification, history of ART, CD4 cell count and percentage and plasma HIV RNA measurements during receipt of NNRTI-based highly active antiretroviral therapy (HAART) and prior to switching to PI-based HAART, and genotypic resistance test results before switching to PI. During the period under study, viral load was not monitored routinely, but generally tested at the time of clinical or immunological failure. The study was approved by the Institutional Review Boards of all sites.

The interpretation of mutations was based on the guidelines published by the International AIDS Society (IAS)-USA Drug Resistance Mutations group [13]. For this study, NRTI resistance mutations included M41L, D67N, K70R, L210W, T215F/Y, and K219Q/E thymidine analogue-associated mutations (TAMs), the Q151M complex, the 69 insertion complex, K65R, L74V, K70E, Y115F and M184V/I. Multi-NRTI resistance was defined as having at least four TAMs or the presence of Q151M or the 69 insertion. NNRTI-associated mutations included V90I, A98G, L100I, K101E/H/P, K103N, V106A/M, V108I, E138A, V179D/F/T, Y181C/I/V, Y188C/L/H, G190S/A, P225H and M230L. The etravirine-weighted mutation score was calculated according to the importance of the mutations [14]. Four mutations merited a weighting factor of 4: L100I, K101P and Y181C/I. Mutations with a weighting factor of 3 were E138A/G, V179E, G190Q, M230L and K238N. Weighting scores of 2 were assigned to K101E, V106A, E138K, V179L and Y188L, while mutations at 11 sites had a score of 1: V90I, K101H, V106M, E138Q, V179D/F/M, Y181F, V189I, G190E/T, H221Y, P225H and K238T. A weight mutation score of ≥ 4 was interpreted as being associated with a significant reduction in etravirine efficacy [12]. Genotypic resistance testing was performed using the TruGene HIV-1 Genotyping system (Visible Genetics, Inc., Toronto, Canada) at five sites, the ViroSeq HIV-1 Genotyping System (Celera Diagnostics, Alameda, CA) at one site, and an in-house method using Stanford and IAS databases [15] at two sites.

Statistical analysis

Descriptive analyses were performed to describe baseline patient characteristics, using median (interquartile range) and frequencies as appropriate. The proportions of patients with various NRTI- and NNRTI-associated mutations were

determined. Associations between NNRTI-based regimens and types of mutation were evaluated using χ^2 and Fisher's exact tests as appropriate. Factors including the duration of the NNRTI-based regimen used (per year), the CD4 percentage (categorized as $<$ and $\geq 15\%$), and the plasma HIV RNA level (categorized as plasma HIV RNA $>$ and $\leq 5 \log_{10}$ copies/mL) at the time of genotypic resistance testing were examined for associations with multi-NRTI resistance and etravirine resistance using univariate and multivariate logistic regression analysis. Mean and median numbers of NNRTI mutations in efavirenz- and nevirapine-exposed children were compared using Student's *t*-test and the Wilcoxon rank sum test. Ninety-five per cent confidence intervals (CIs) were calculated by Wald-based *P*-values, and $P < 0.05$ was considered statistically significant. Analyses were performed using SAS version 9.1 (SAS Institute, Cary, NC, USA).

Results

Between September 2002 and June 2007, there were 151 children who met the inclusion criteria of experiencing failure of an NNRTI-based regimen and requiring a treatment switch to a second-line PI-based regimen. Genotype testing results were obtained for 120 children (79%). The other 31 children did not have genotype testing performed prior to switch and did not have stored plasma available. The data were transferred from clinical sites to the data management centre from December 2007 to August 2008. Baseline characteristics at initiation of first-line regimens are presented in Table 1. Patients suffered severe immunodeficiency prior to initiation of ART, as demonstrated by their advanced CDC stages and low CD4 levels. The majority of children were on stavudine, lamivudine and nevirapine. The median duration of NNRTI-based regimens prior to genotype testing was 23.7 months. There was no difference in duration of treatment between children who experienced failure of nevirapine- and efavirenz-based regimens ($P = 0.75$). The median CD4 percentage and HIV RNA at the time of genotyping were 12% and $4.8 \log_{10}$ copies/mL, respectively. Treatment failure was documented as clinical failure in 38 children (32%), immunological failure in 47 children (39%), and unspecified in 35 children (29%).

Genotypic drug resistance test

The frequencies of selected mutations in the reverse transcriptase gene are shown in Tables 2 and 3. The most commonly detected mutation was M184V/I (85%) for lamivudine resistance. The prevalences of multi-NRTI-associated mutations were 22.5% for at least four TAMs, 11.7% for the Q151M complex and 1% for the 69 insertion. In the

Table 1 Baseline characteristics of 120 HIV-infected children who experienced failure of nonnucleoside reverse transcriptase inhibitor-based antiretroviral therapy (ART)

Clinical characteristics	Results
Patient characteristics at initiation of ART	
Age at initiation of ART (years) [median (IQR)]	6.8 (4.6–9.2)
Male [<i>n</i> (%)]	65 (54)
Clinical characteristics prior to ART initiation	
CDC clinical classification N:A:B:C [<i>n</i> (%)]	5:30:53:32 (4:25:44:27)
CD4 percentage [median (IQR)]	3 (1–8)
CD4 cell count (cells/ μ L) [median (IQR)]	66 (20–242)
Plasma HIV RNA (log ₁₀ copies/mL) [median (IQR)]	5.3 (4.7–5.9)
First antiretroviral drug regimen	
NNRTI [<i>n</i> (%)]	
Nevirapine	91 (76)
Efavirenz	29 (24)
NRTI [<i>n</i> (%)]	
Stavudine/lamivudine	88 (73)
Zidovudine/lamivudine	32 (27)
Patient characteristics at time of resistance testing	
Duration of NNRTI-based therapy (months) [median (IQR)]	23.7 (15.7–32.6)
CD4 percentage [median (IQR)]	12 (4–20)
CD4 count (cells/ μ L) [median (IQR)]	211 (93–582)
Plasma HIV RNA* (log ₁₀ copies/mL) [median (IQR)]	4.8 (4.3–5.2)

*Plasma HIV RNA was available for 116 children.

ART, antiretroviral therapy; CDC, Centers for Disease Control and Prevention; IQR, interquartile range; NNRTI, nonnucleoside reverse transcriptase inhibitor; NRTI, nucleoside reverse transcriptase inhibitor.

Table 2 Frequency of selected resistance mutations in the reverse transcriptase gene related to nucleoside reverse transcriptase inhibitors (NRTIs) in 120 HIV-infected children

Genotypic resistance mutation	Number (%)
NRTI	
Any NRTI	118 (98.3)
M184V/I	102 (85.0)
Q151M complex	14 (11.7)
K65R	6 (5.0)
Insertion 69	1 (0.8)
TAMs	
M41L	29 (24.2)
D67N	48 (40.0)
K70R	24 (20.0)
L210W	21 (17.5)
T215F/Y	51 (42.5)
K219Q/E	25 (20.8)
Any TAM	73 (60.8)
1–3 TAMs	46 (63.0)
≥ 4 TAMs	27 (37.0)
K70E	1 (0.8)
L74V	7 (5.8)
Y115F	2 (1.7)

TAM, thymidine analogue mutation.

multivariate analysis, the predictors of multi-NRTI resistance were CD4 < 15% prior to switching regimen, with an odds ratio (OR) of 5.49 (95% CI 2.02–14.93) and plasma HIV RNA > 5 log₁₀ copies/mL, with an OR of 2.46 (95% CI 1.04–5.82) (Table 4).

Table 3 Frequency of selected resistance mutations in the reverse transcriptase gene related to nonnucleoside reverse transcriptase inhibitors (NNRTIs) in 120 HIV-infected children

	EFV-based (<i>n</i> = 29)	NVP-based (<i>n</i> = 91)	Total (<i>n</i> = 120)	<i>P</i> -value
Any NNRTI mutation [<i>n</i> (%)]	28 (96.6)	89 (97.8)	117 (97.5)	0.57
Genotypic resistance mutation				
L100I	3 (10.3)	0 (0)	3 (2.5)	1.00
K103N	22 (75.9)	20 (22)	42 (35)	< 0.001
V106A	1 (3.5)	0 (0)	1 (0.8)	1.00
V106M	0 (0)	1 (1.1)	1 (0.8)	0.76
V108I	6 (20.7)	14 (15.4)	20 (16.7)	0.83
Y181C	5 (17.2)	47 (51.7)	52 (43.3)	0.001
Y181I	0 (0)	1 (1.1)	1 (0.8)	0.76
Y188C	0 (0)	2 (2.2)	2 (1.7)	0.57
Y188H	2 (6.9)	0 (0)	2 (1.7)	1.00
Y188L	6 (20.7)	7 (7.7)	13 (10.8)	0.99
G190S	1 (3.5)	2 (2.2)	3 (2.5)	0.86
G190A	7 (24.1)	27 (29.7)	34 (28.3)	0.57
Etravirine mutation scores*				
No mutation	14 (48.3)	24 (26.4)	38 (31.7)	0.12
1–2	6 (20.7)	17 (18.7)	23 (19.2)	
2.5–3.5	0 (0)	2 (2.2)	2 (2.4)	
≥ 4	9 (31.0)	48 (52.8)	57 (47.5)	

Values in bold denote *P*-value < 0.05.

Only major NNRTI mutations are shown.

*Etravirine mutation scores are based on Madruga *et al.* [12].

EFV, efavirenz; NVP, nevirapine.

The most common NNRTI mutations were Y181C/I (44%), K103N (35%) and G190A/S (31%). The K103N mutation was more common in children who received efavirenz than in those who received nevirapine ($P < 0.001$), while Y181C was more common among children who received nevirapine ($P = 0.001$). Forty-eight per cent of children had etravirine mutation-weighted scores ≥ 4 . There was a trend towards a higher rate of etravirine mutation scores ≥ 4 among children who received nevirapine than among those on efavirenz (52.8% *vs.* 31.0%; $P = 0.12$). In the univariate analysis, there was no association between the duration of NNRTI treatment, the CD4 percentage, or plasma HIV RNA and the risk of etravirine resistance.

Discussion

This study investigated the HIV resistance pattern in children with treatment failure on WHO-recommended first-line NNRTI-based ART. Eighty-five per cent of the children had resistance to lamivudine, and about a quarter of the children had multi-NRTI resistance mutations conferring resistance to all NRTI drugs, which limit opportunities for recycling NRTIs as a component of the second-line PI-based regimen. Ninety-eight per cent of the children had at least one mutation related to NNRTIs, with half having high-grade etravirine resistance. A CD4 percentage < 15% and an HIV RNA > 5 log₁₀ copies/mL at

Table 4 Predictive factors of multiple nucleoside reverse transcriptase inhibitor (NRTI) resistance among children who experienced failure of nonnucleoside reverse transcriptase inhibitor (NNRTI)-based antiretroviral therapy (ART)

		Univariate		Multivariate	
Factor	Multiple NRTI resistance [n/N (%)]	OR (95% CI)	P-value	OR (95% CI)	P-value
Duration of NNRTI-based ART (years)					
1	0/4	–	–	–	
2	16/45 (35.6)	1		–	
3	13/42 (31.0)	0.81 (0.33–1.99)	0.649	–	
> 3	15/29 (51.7)	1.94 (0.75–5.02)	0.171	–	
HIV RNA prior to switch to PI (log ₁₀ copies/mL)					
≤ 5	4/23 (17.4)	1		1	
> 5	38/93 (40.9)	3.42 (1.53–7.66)	0.003	2.46 (1.04–5.82)	0.040
CD4 percentage prior to switch to PI					
< 15%	38/75 (50.7)	6.68 (2.53–17.64)	< 0.001	5.49 (2.02–14.93)	0.001
≥ 15%	6/45 (13.3)	1		1	

Values in bold denote *P*-value < 0.05.

Multi-NRTI resistance was defined as having at least four thymidine analogue mutations (TAMs) or the presence of the Q151M complex or insertion 69. CI, confidence interval; OR, odds ratio; PI, protease inhibitor.

the time of genotype testing predicted multi-NRTI resistance.

First-line NNRTI-based treatment failure is a major public health problem, especially in children, because of the limited availability of approved second-line antiretroviral drugs and access to new drugs. Moreover, the lack of routine viral load monitoring in many resource-limited countries leads to delay in early detection of children who have virological failure. This causes accumulation of mutations within the NRTI and NNRTI drug classes until treatment failure is finally diagnosed on the basis of clinical or immunological criteria [16]. Lapphra *et al.* reported that 8.4% of Thai children who started NNRTI regimens had treatment failure at 24 months [17]. Jittamala *et al.* [18] recently showed that 20% of Thai children had virological failure within 5 years of starting NNRTI-based regimens, with the majority failing in the first 12 months. These reports underscore the need for an understanding of resistance development, in order to design effective second-line regimens, especially if the availability of genotype testing is limited. Recently, the National Health Security Office, which provides ART to almost all HIV-infected Thai children, reported that 20% of HIV-infected Thai children are receiving second-line PI regimens. The regional Asian network, Treat Asia, which follows over 1000 children, also reported that 20% of children were on second-line ART [19].

The children in our study were from eight large paediatric HIV centres in Thailand. Similar to other studies on children from South Africa [6] and Thailand [8,18], extensive NRTI mutations were found. The rate of multi-NRTI resistance with at least four TAMs was as high as 23%, which limits the potential for recycling of NRTIs,

including tenofovir. A reduction in the efficacy of tenofovir has been reported in the presence of the K65R mutation or at least three TAMs inclusive of M41L or L210 [20]. The prevalence of K65R was only 5%, which is similar to that in Thai adults [9], but lower than the 15% reported in South African children [6], and the 23% reported in Malawian adults [21], which could be explained by differences in HIV subtypes or duration of treatment. There was no difference in the frequency of K65R between children who received zidovudine (6.3%) and those who received stavudine (4.5%). Tenofovir is currently approved for adults, and the results from randomized trials in children will be available in 2010. As tenofovir is usually the only NRTI with some antiviral activity that remains an option in children with multi-NRTI resistance, the ability to use tenofovir in second-line regimens will increase the odds of viral suppression.

We found that around 98% of children had NNRTI resistance mutations that would render nevirapine and efavirenz ineffective. As previously reported, Y181C was mostly associated with nevirapine failure, while K103N was associated with efavirenz failure [22]. Etravirine is a new drug in the NNRTI class that continues to have antiviral activity after the development of a few NNRTI mutations, especially if those mutations do not include Y181C [11]. Using a weighted scoring system for assessing etravirine resistance, [14] we found that almost half of our children had high-grade etravirine resistance, which was higher than the proportion found in other reports in children [23] and in adults [24]. Etravirine has been used successfully in adults with multi-class failure as an alternative to PI-based salvage regimens [12]. It is not yet approved in children, but studies are ongoing to evaluate the efficacy of this drug in the

setting of triple class failure. Our data show that the opportunity to use efavirenz in late NNRTI failure is limited because of the high rates of high-grade efavirenz resistance.

In this study, high viral load was a predictor of multi-NRTI resistance, which is similar to results from a Thai adult study [9]. A CD4 percentage of <15% at the time of failure also predicted multi-NRTI resistance. Similarly, in a study of Malawian adults who failed first-line ART, patients with CD4 counts <100 cells/ μ L had a 6.1-times higher risk of harbouring the K65R resistance mutation [21]. Similar to a report in Thai adults, we could not find predictors of high-grade efavirenz resistance [24]. Among four children who developed treatment failure within the first year of treatment, there was no multi-NRTI resistance. This suggests that early viral load monitoring, at least during the first year of treatment, could aid the detection of early failure and the design of an optimal second-line regimen. Gupta *et al.* conducted a meta-analysis in an HIV-infected adult population, which showed that patients treated in settings with virological monitoring less frequent than every 3 months had a higher risk of NNRTI resistance, TAMs, and lamivudine resistance [25]. Recently, the Thai national programme began to provide yearly viral load monitoring. There are ongoing efforts to make viral load monitoring feasible in resource-limited settings, for example using the dried blood spots technique [26].

Our study has several limitations. Firstly, its retrospective design could have resulted in incomplete data collection and failure to include children who died before switching to second-line therapy; however, this kind of bias would probably have led to an underestimation of the impact of drug resistance. Secondly, the population in this study was at an advanced disease stage, with very low baseline CD4 percentages prior to ART initiation and at the time of treatment switch, which may have resulted in bias towards high rates of multi-drug resistance. However, this reflects real life situations in most resource-limited settings where treatment failure is usually detected when patients experience immunological or clinical failure. Thirdly, all the sites involved in this study followed the practice guidelines set by the Thai Ministry of Public Health by having CD4 monitoring at least every 6 months, and having viral load measurements performed only when patients met the criteria for immunological or clinical failure. Therefore, we do not have information on the duration of virological failure prior to the genotypic resistance testing. However, we used the duration of the NNRTI-based regimen as a surrogate marker for the analysis of the predictors of multi-drug resistance.

In summary, in children who did not have access to routine viral load monitoring and who experienced failure of WHO-recommended first-line NNRTI therapy, there were

high rates of lamivudine, nevirapine and efavirenz resistance. Multi-NRTI resistance was found in a quarter of patients and almost half had high-grade efavirenz resistance. Therefore, the appropriate second-line regimen is a boosted PI-based regimen, with a limited role for efavirenz. Further studies should be carried out to determine whether routine viral load monitoring for children would reduce the rate of multi-drug resistance and have any additional benefit in improving outcomes of second-line regimens in HIV-infected children living in resource-limited settings.

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